

G D HRUTHIK

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years of experience in designing and deploying end-to-end ML pipelines and scalable production solutions. Expert in TensorFlow, PyTorch, and MLOps, with a track record of improving product performance in clinical, industrial, and e-commerce environments. Recognized for transforming research models into reliable, measurable business impacts through innovative AI solutions.

EDUCATION

New England College, Henniker, NH
Master of Science, Data Science and Analytics
• **GPA:** 3.74

Jun 2025

PROFESSIONAL EXPERIENCE

Sanofi

Machine Learning Engineer

Apr 2024 - Present

Boston, MA

- Collaborated with clinical teams to establish end-to-end ML pipelines for genomic and clinical trial data, leveraging Python and R for exploratory data analysis and model prototyping.
- Containerized models with Docker and deployed them via Kubernetes to enable scalable production pipelines for oncology and immunology studies.
- Instituted robust model monitoring, versioning, and compliance processes in line with FDA and GxP requirements, ensuring operational excellence for ML systems.
- Developed internal AI tools using Streamlit and FastAPI that enhanced cross-team collaboration and increased adoption of AI solutions among scientists by 40%.
- Integrated EHR and biomarker data into production pipelines, streamlining data analysis and supporting translational medicine teams in biomarker discovery.
- Engineered a computer vision workflow with CNNs and PyTorch for microscopy images, reducing protein misfolding detection time by 50% compared to manual review.

Cognizant Technology Services

Machine Learning Engineer

Aug 2021 - May 2023

Coimbatore, TN, India

- Designed and prototyped real-time anomaly detection models using Autoencoders and Isolation Forests, contributing to production-level ML solutions that cut equipment downtime by 20%.
- Built and optimized predictive maintenance pipelines using Random Forest integrated with Kafka and Azure Stream Analytics, ensuring streamlined data pipelines and scalable inference.
- Deployed a fraud detection solution with Azure ML, achieving sub-120ms inference speeds and setting benchmarks for model operational excellence and automated monitoring.
- Supported an e-commerce recommendation system in TensorFlow that led to a 15% increase in click-through rates during A/B testing, encouraging innovation in ML-driven product enhancements.
- Standardized end-to-end ML operations pipelines through containerization and integrated monitoring, reinforcing best practices for production ML model deployment and performance tracking.

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python, C++, R, SQL, Scala, Pandas, NumPy
- **Machine Learning:** Regression, Classification, Clustering, Dimensionality Reduction, Ensemble Methods (Bagging, Boosting, Stacking), Hyperparameter Tuning (Optuna, Grid Search)
- **Deep Learning & Computer Vision:** CNN, RNN, LSTM, BERT, CodonBert, ProGen2 Transformers, Attention Mechanisms, Reinforcement Learning, CLIP, Image Classification, Object Detection, Fine-tuning Models, Diffusion Models
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, DeepSpeed, Flask, Django
- **Natural Language Processing:** Text Preprocessing, Tokenization, Word Embeddings, Sentiment Analysis, LangChain, RAG, Hugging Face Transformers, OpenAI API
- **Generative AI:** GPT-based LLMs, Diffusion Models, Prompt Engineering
- **Vector Search & Retrieval:** FAISS, Pinecone, Chroma, Weaviate
- **MLOps & Model Deployment:** MLflow, DVC, Docker, Kubernetes, Azure Kubernetes Service (AKS), Streamlit, Ray, TorchServe, FastAPI, Triton Inference Server, TensorFlow Serving
- **CI/CD & Workflow Orchestration:** Jenkins, GitHub Actions, Apache Airflow, Azure DevOps, Luigi
- **Cloud Platforms:** AWS SageMaker, GCP Vertex AI, Azure ML Studio
- **Visualization & Reporting:** Tableau, Matplotlib, Seaborn
- **Data Engineering:** Hadoop, PySpark, Azure Databricks, Data Factory, Synapse Analytics, Snowflake
- **Project & Collaboration Tools:** Jupyter, Google Colab, VS Code, Agile, A/B Testing, Requirement Gathering